

Evaluating Name-Only Directory Routing for One-Shot Code Search

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Abstract

Coding agents must find a small set of relevant files in large repositories. Lexical tools are fast but may rank incidental word matches above the files needed to understand an issue. Repository folders and file names provide a different signal: a model can navigate this hierarchy and nominate candidate files without an embedding index. We study when model-guided directory routing retrieves annotated code files that fixed-query lexical rankings miss, and whether their combination improves evidence delivered under a fixed context budget. We compare path-name tree routing with actual `rg`, FTS5, and equal-budget fusion on 82 audited issues from 11 repositories at pinned pre-fix commits. At eight files, tree routing recovered 0.465 of annotated gold files on average, versus 0.352 for FTS5 and 0.245 for a fixed full-issue `rg` query. The paired tree-FTS5 difference was 0.113 (95% repository-cluster bootstrap interval 0.053–0.168). Tree found 61 gold-file occurrences absent from FTS5’s top eight across 40 issues. Fusion improved recall over FTS5 to 0.491, but its 0.026 gain over tree alone had an interval crossing zero; at 16 files, fusion recovered fewer gold files than tree. Under a shared 16K-token context selector, tree delivered 0.443 of annotated lines versus 0.246 for FTS5 on the 55 cases whose annotation text aligns with pre-fix source. A post-review flat path tournament using the same model reached 0.572 file recall@8, above tree, but used 24.6 versus 8.9 model calls per issue. Tree routing averaged 8.9 seconds and 12.2K input tokens per issue, versus a 0.9-second FTS5 build and 7-millisecond query. These results support model-guided path-name search as a complementary candidate source for one-shot retrieval; they do not establish a quality advantage from hierarchy or superiority over an adaptive coding agent.

1 Introduction

A coding agent can call `rg` repeatedly, but the first query is often a behavioral description rather than an identifier. Search terms may match documentation, changelogs, tests, and unrelated implementation. The repository’s directory structure supplies an alternate route to candidate files: at each directory, a model selects plausible children and may retain several branches. This paper asks whether that route discovers *additional annotated source evidence* at a useful cost, not whether it makes text search obsolete. Our contribution is a controlled, reproducible evaluation of when a language-model-guided walk of repository names adds files omitted by fixed lexical rankings, and whether those files survive an equal context-selection budget. We do not introduce hierarchical code search.

The claim has two stages. First, a file-discovery method must nominate gold files within a fixed candidate limit. Second, those candidates must yield the required source lines within the coding agent’s context budget. A method can win the first stage and lose the second through ordering, overly narrow windows, or noisy context. We therefore report both stages rather than using span overlap or downstream answer scores as a proxy for file discovery.

Our pilot motivates the question but does not answer it generally. On 24 previously inspected issues from Django and Svelte, name-only tree routing found a larger fraction of gold files at eight candidates than one fixed full-issue `rg` literal ranking (0.476 versus 0.219). A stronger FTS5 control reached 0.295; tree and FTS5 both delivered complete gold-line contexts for only 4/24 questions at 16K tokens. These observations demand a new-repository held-out comparison with multiple lexical controls and a shared final-context budget. We make no inference from a selected three-file pilot example to general performance.

2 Research questions

- **RQ1:** When does model-guided directory routing add gold files that tuned lexical search does not rank within the same top-k budget?
- **RQ2:** Does adding tree candidates to lexical candidates improve complete file and line evidence under equal candidate and context budgets?

3 Method and evaluation

3.1 Task and data

The unit is an issue statement paired with a repository at its pre-fix base commit. The retriever receives the issue text and source tree only: never annotated paths, gold line ranges, a solution patch, or post-fix code. ContextBench [3] supplies repository issues and annotated source regions. We pinned its revision and Parquet checksum in a `evals/fixtures/queries/contextbench_paper.json`. The 24 Django/Svelte pilot cases are excluded. The independent manifest contains 87 issues at 87 commits from 11 other repositories in Python, TypeScript, JavaScript, Go, C, and Rust. Repositories were selected by eligible-case count within language strata; at most eight issues per repository were selected by a hash ordering of instance ID. This tests transfer across repositories but is not a random sample of software projects. Source availability and common-corpus eligibility are audited before scoring, with exclusions logged and never replaced. The exact selection and audit rules are in the accompanying `paper/study-amendment-001.md`. The source audit retained 82/87 cases. All five exclusions had gold paths outside the common searchable-file policy, including real source under directories named `build`. They remain in the exclusion ledger. Among the retained cases, 55 had every annotated span text align with the base-commit source after whitespace normalization or exact-text containment. File discovery uses all 82; line-level conclusions are shown both for all 82 and the 55-case aligned-text sensitivity subset. The initial 171-case tree pass was stopped at the author’s request after the reduced subset had already completed. Some interim tree outcomes were visible before the budget amendment, but the first-eight-per-repository rule depends only on the original hash-ordered manifest, not those outcomes. The study is not externally preregistered or fully blinded.

Cohort by language, before and after the source audit.

Language	Selected	Audited
Python	24	24
TypeScript	16	16
JavaScript	16	16
Go	16	13
C	7	7
Rust	8	6
Total	87	82

3.2 Retrieval arms

The tree arm walks the existing folder hierarchy. At each visited directory, a categorical classifier receives the first 500 issue characters and a menu of immediate child directory and file names. Each directory option also includes up to 12 descendant names from at most three levels below it. The classifier assigns probabilities and may keep multiple plausible branches. The evaluated implementation uses `typesafe/jev-1.13.0`, no generated directory summaries, minimum and relative branch-probability thresholds of 0.01, a 32-call cap, and a 180-second per-issue time cap. It outputs at most 16 ranked file paths. This is a file-discovery method, not a complete coding agent.

Here “classifier” means the TypeSafe System One `choice` endpoint, not a classifier trained for this study. The request state is `User question:` followed by the first 500 issue characters. Each criterion is an immediate child path and name-only summary; directory criteria additionally list up to 12 descendant names.

A `none_of_these` criterion is added. The instruction is: “Choose the best route toward answer evidence. Directory options summarize descendants, so select a directory when a descendant could answer. Choose none only when the question is unrelated to every option.” The provider returns a probability for every criterion. The JSON request has `questions.route.type=choice`, `instructions`, and `criteria={option_i: text,..., none_of_these: text}`; the adapter reads `answers.route.probabilities` and rejects missing or out-of-range values. We do not renormalize probabilities across menus. Menus are split at 254 options or 48,000 request characters; each option is limited to 800 characters. The `none_of_these` probability is not a traversable child; the best-child threshold excludes it. Children with probability at least the larger of 0.01 and 0.01 times the best child probability are retained. A stack visits higher-scoring children first; tied children follow reverse input order. Path rank is traversal order; branch probabilities are not multiplied into a global file score. Single-child directories need no model call. The API request supplies no temperature setting. HTTP 429/529 responses are retried up to three times; other errors fail the issue. The archived model identifier is `jev-1.13.0`; provider-side weights and sampling behavior are not under our control.

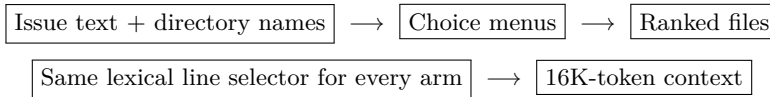


Figure 1: Directory routing proposes files; the shared selector and packer determine which source lines reach the context budget.

Lexical arms search the same eligible UTF-8 files. The primary comparator is in-memory SQLite FTS5 ranking from the first 500 issue characters, chosen because it had the highest lexical recall@8 on the pilot. It indexes paths and file content, searches an OR of unique non-stopword tokens, and ranks with FTS5’s BM25 function using path/content weights 2/1. An actual full-issue ripgrep replay searches case-insensitive literal terms over the same files; its fixed file ranker weights term rarity and path matches. It is one nonadaptive query, not a proxy for an agent that can run successive searches or generate better keywords. The hybrid applies reciprocal-rank fusion (rank constant 60) to tree and FTS5 top-16 lists and truncates back to eight or sixteen files; it receives no larger candidate allowance. We do not compare its one-shot file ranking against Pi’s multi-turn `grep/read` loop using span recall; that would measure different systems and conflate file discovery with section selection.

3.3 Outcomes and statistics

For issue q , let G be its annotated gold-file set and $R(k)$ the first k returned files. File recall is $|G \cap R(k)| / |G|$; the main aggregate is the unweighted mean over issues. We also count fully recovered gold-file sets, gold files found by tree but absent from the same-size FTS5 prefix, and issues with at least one such file. Since 16 of 82 audited issues have more than eight gold files, complete recovery at eight candidates is impossible for them; the ceiling is reported. File precision is secondary because the intended use tolerates some extra files if required evidence is present.

Candidate files are not equivalent to delivered context. Every ranking uses the same line selector: at most four lexical anchors per file, a 200-line radius around each, and merged overlapping windows. The windows are packed under a 16K-token budget with the same tokenizer and order. The selector tokenizes the full issue and every retrieved file, removes stopwords and tokens of at most two characters, and scores each source line by its count of distinct overlapping query terms. It takes the four highest-scoring lines per file (earlier line wins ties), expands each by 200 lines in both directions, and merges touching ranges. When a file has no matching line, it offers the whole file. Ranges are packed in file-rank order, with higher-scoring ranges first within a file; an over-budget range is skipped and packing continues. The tokenizer is `cl100k_base`, and line numbers and paths count toward the budget. The selector scans full retrieved files, but receives only the issue text and those files. Gold paths, annotated line text/ranges, patches, and post-fix code are never passed to retrieval or selection; they enter only the subsequent scorer. The same selector is rerun for every arm without arm-specific settings.

```

for file in ranked_files:
    hits = top_four_matching_lines(issue_terms, full_file)
    ranges = merge(expand(hits, radius=200)) if hits else [full_file]
    for range in highest_hit_score_first(ranges):
        if used_tokens + numbered_block_tokens(range) <= 16000:
            include(range)

```

We score annotated-line recall and complete line coverage after packing. Because text annotations can differ from base-commit source, we show line outcomes separately on the pre-retrieval aligned-text subset. Paired differences use 10,000 repository-cluster bootstrap resamples (seed zero): sample the 11 repositories with replacement and carry all their paired issues into each sample. Per-repository results expose concentrated gains. Setup, query, model-call, token, cost, and wall-time measurements are reported separately. Neither file nor line recall is issue-resolution success. The reduced study does not repeat the tree run or measure the combined hybrid pipeline wall time; it reports each measured arm’s operations and does not claim a fused end-to-end speedup. Prespecified descriptive subgroups mark whether the issue explicitly mentions an annotated full path or basename and whether its gold files span more than one directory. These labels use the issue and gold metadata, never retrieval outputs.

4 Results

4.1 Equal-budget retrieval

Table 1 reports unweighted issue means. Every arm uses the same candidate limit and context selector. The line-recall columns score annotated lines actually included after the 16K-token pack on the 55 cases with aligned annotation text, not just lines in nominated files. File columns use all 82 cases. Complete-file rates have ceilings of 66/82 at eight candidates and 80/82 at sixteen because some issues annotate more files than the limit.

Table 1: Equal-budget retrieval and delivered-line results. File values use all 82 issues; L uses the 55 fully aligned-text issues. All values are issue means. Full counts issues with every gold file in the top- k . Fusion uses equal- k RRF with rank constant 60.

Arm	File@8	Full@8	L55@8	File@16	Full@16	L55@16
Fixed full-issue rg	0.245	12/82	0.116	0.328	16/82	0.116
FTS5, first 500 chars	0.352	17/82	0.246	0.421	19/82	0.252
Name-only tree	0.465	25/82	0.443	0.591	32/82	0.457
Tree + FTS5	0.491	23/82	0.499	0.571	30/82	0.504

At eight candidates, tree had higher file recall than FTS5 on 33 issues, lower on 15, and equal on 34. Its 61 gold-file occurrences absent from FTS5’s top eight appeared in 40 issues; FTS5 had 34 gold-file occurrences absent from tree’s top eight. Against fixed **rg**, tree had 82 such occurrences in 49 issues and **rg** had 29 absent from tree. A further descriptive overlap count found 57 tree hits across 37 issues absent from both lexical top-eight lists. These are *issue-file occurrences*, not distinct files across the corpus. Tree alone recovered eight more complete gold-file sets than FTS5 at top eight. The fusion’s higher mean recall did not translate to more complete sets than tree: it recovered 23 versus 25. The primary macro average weights issues equally. A secondary micro average weights each annotated gold-file occurrence equally: at eight candidates, tree found 108/391 (0.276) and FTS5 found 81/391 (0.207); fusion found 120/391 (0.307). At sixteen, tree found 150/391 (0.384) and fusion 151/391 (0.386). Thus the small tree–fusion ordering at sixteen changes with the aggregation unit, while tree exceeds FTS5 under both.

Table 2 gives prespecified paired differences; intervals resample the 11 repositories, retaining all issues within each sampled repository. A positive value favors the left arm. The intervals describe sampling uncertainty across this convenience cohort, not a general population guarantee. A post-study exact sign-flip sensitivity over the 11 repository-level paired tree-FTS5 contrasts gave $p = 0.0078$ at eight files under exchangeable repository signs; it is not a test of generalization to unsampled repositories.

Table 2: Paired issue-mean differences and 95% repository-cluster bootstrap intervals. File columns use 82 cases; Line@8 uses the 55 aligned-text cases. Positive values favor the left arm.

Comparison	File@8	Line@8	File@16
Tree – FTS5	+0.113 [0.053, 0.168]	+0.196 [0.125, 0.275]	+0.170 [0.085, 0.252]
Tree – fixed rg	+0.220 [0.106, 0.342]	+0.326 [0.242, 0.415]	+0.263 [0.148, 0.394]
Fusion – FTS5	+0.139 [0.092, 0.188]	+0.252 [0.181, 0.340]	+0.150 [0.089, 0.219]
Fusion – tree	+0.026 [-0.013, 0.069]	+0.056 [0.002, 0.126]	-0.020 [-0.056, 0.018]

Fusion therefore improves this FTS5 ranking, but is not a demonstrated improvement over tree alone. At top eight it raises file recall in 20 issues, lowers it in 13, and leaves it unchanged in 49 relative to tree. At top sixteen, fusion lowers mean candidate recall slightly. Rank fusion can displace a tree-selected gold file at a fixed limit; combining candidate lists is not free. The aligned-text line difference for fusion versus tree is +0.056 [0.002, 0.126]; this pre-retrieval aligned subset is smaller and easier on file recall, so it does not establish a robust general gain. On all 82 issues the annotation-sensitive difference is +0.043, interval -0.006 to +0.101.

4.2 Delivered context and annotation sensitivity

The selector packed, on average, 12.7K tokens for tree@8, 14.2K for FTS5@8, 14.9K for **rg**@8, and 13.8K for fusion@8, all within the same 16K cap. Greater tree line recall was not explained by a larger context allowance. Complete annotated-line coverage was nevertheless uncommon: 14/82 issues for tree, 8/82 for FTS5, 5/82 for **rg**, and 15/82 for fusion. File discovery remains only the first step toward sufficient coding context.

For the 55 issues whose every annotated text span aligns to the pinned pre-fix source, line recall@8 was 0.443 for tree, 0.246 for FTS5, 0.116 for **rg**, and 0.499 for fusion. The paired tree-FTS5 difference was +0.196 [0.125, 0.275], and fusion-tree was +0.056 [0.002, 0.126]. These are the most trustworthy exact-line comparisons. The predeclared all-82 line analysis remains in the archived artifact, but is annotation-sensitive: 940/991 spans aligned with the pinned source and 51 spans across 27 issues did not. A post-study mechanical audit found the annotated text elsewhere in the same file for 20 mismatched spans and did not find it in that file for 31; these labels do not establish why the dataset and source differ. The mismatch is fixed before retrieval and cannot vary by retrieval arm. Mismatched-case counts range from zero in GitHub CLI, Transformers, and PonyC to six of eight in Material UI; every repository’s count and case IDs are archived with the audit. We do not infer an exact-line improvement on the 27 cases from the numeric range overlap alone. The aligned subset is not a random sample of difficulty: tree file recall@8 was 0.531 there versus 0.331 on the 27 mismatched cases. We use it for annotation validity, not as a representative replacement for all 82 issues.

Table 3: Mechanical alignment audit of 991 annotated spans. Fifty-five issues align completely; 27 have at least one mismatch. The mismatch categories are text-location checks, not cause labels.

Span result	Count
Aligned at annotated range	940
Annotated text elsewhere in same file	20
Annotated text not found in same file	31

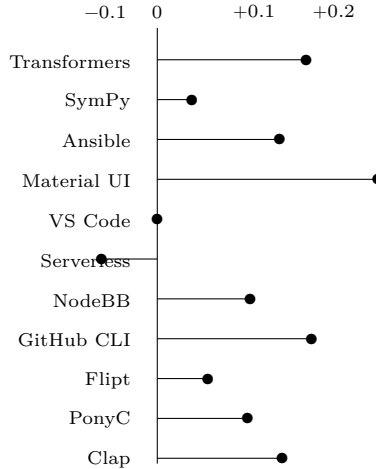


Figure 2: Repository mean difference in file recall@8 (tree minus FTS5). Dots right of zero favor tree; each repository contributes 5–8 issues.

4.3 Repository variation and operational cost

Tree exceeded FTS5 file recall@8 in nine of the 11 repositories, tied in VS Code (0.13 each), and trailed in Serverless (0.16 versus 0.23). The advantage was small for multi-directory gold sets (47 issues, +0.027 tree–FTS5) compared with single-directory sets (35 issues, +0.229). This subgroup pattern is descriptive and may reflect both repository mix and the difficulty of preserving several branches. It does not isolate hierarchy as the cause of the gain. Nine issues explicitly name a gold path; tree–FTS5 recall gain is +0.078 there versus +0.117 in the other 73. These small path-hint subgroups also do not establish a mechanism. Table 4 shows the repository-level means, making both improvements and exceptions visible.

Table 4: File recall at eight candidates by repository.

Repository	n	Tree recall@8	FTS5 recall@8	Fusion recall@8
transformers	8	0.78	0.59	0.78
SymPy	8	0.69	0.65	0.73
Ansible	8	0.60	0.45	0.57
Material UI	8	0.61	0.34	0.55
VS Code	8	0.13	0.13	0.13
Serverless	8	0.16	0.23	0.28
NodeBB	8	0.18	0.07	0.14
GitHub CLI	8	0.66	0.47	0.63
Flit	5	0.29	0.23	0.42
PonyC	7	0.42	0.31	0.55
Clap	6	0.54	0.38	0.66

The tree made 729 Choice-model calls over 82 issues (8.89 per issue), averaging 12.2K input and 1.93K output tokens, 8.93 seconds wall time, and an estimated \$0.000512 per issue. Its median/p95 routing times were 9.08/13.52 seconds. No tree issue failed or hit the call or time cap.

The estimate multiplies reported input tokens by TypeSafe’s Jev 1.13 rate of \$0.042 per million [4], accessed September 24, 2026; output tokens are free. It is not an observed bill. FTS5 index construction averaged 894 ms per pinned snapshot and its first-500-character query averaged 7.5 ms; **rg** file enumeration averaged 768 ms and full-issue search 1.68 seconds. FTS5 query median/p95 were 4.8/20.6 ms and **rg** search median/p95 were 0.71/6.10 seconds. These are observed operation times, not comparable prebuilt-index versus cold-start end-to-end latency claims. FTS5 and **rg** incurred no model API charge.

Fusion’s end-to-end wall time was not measured; running both sources and packing context cannot be represented by the FTS5 query time alone.

4.4 Post-review path and query controls

After reviewing the original analysis, we added four controls on the same 82 cases. These results are exploratory and did not change the prespecified tree–FTS5 comparison. Path-only FTS5 indexes file names without contents; a basename heuristic counts issue-token overlap in path names. The flat Jev tournament shows every eligible path, in bounded menus, to the same Choice model used by tree routing. Each menu advances its top 16 paths until one ranking remains. It does not use directory menus, but its instruction differs from the tree instruction and it uses more calls. A two-search `rg` script forms a second query from rare terms in the first search’s matched files, then fuses the two lists. It is fixed feedback, not an adaptive coding agent. All four controls use the same file and context budgets and the unchanged line selector.

Table 5: Post-review controls and reference arms. File recall is the issue mean on all 82 cases; L55@8 is delivered-line recall on the 55 cases with aligned annotation text. New controls were not prespecified.

Arm	File@8	File@16	L55@8
Fixed full-issue <code>rg</code>	0.245	0.328	0.116
Two-search <code>rg</code>	0.260	0.316	0.209
Content + path FTS5	0.352	0.421	0.246
Path-only FTS5	0.191	0.237	0.214
Basename overlap	0.173	0.219	0.226
Name-only tree	0.465	0.591	0.443
Flat Jev tournament	0.572	0.630	0.511

Flat Jev exceeded tree at eight files by +0.107 issue-mean recall [0.017, 0.224] in a post-study repository-cluster bootstrap; at sixteen the difference was +0.039 [-0.038, 0.127]. The flat control also delivered more aligned annotated lines at eight files (+0.069 [0.006, 0.130] versus tree), though complete line coverage remained uncommon. Its mean 24.6 calls, 156K input tokens, 30.6 seconds, and estimated \$0.00657 per issue compare with tree’s 8.9 calls, 12.2K input tokens, 8.9 seconds, and estimated \$0.000512. These are observed operations and estimated charges for the completed runs, not a call-matched test of hierarchical traversal. A first timed-out attempt on one large flat case made unarchived partial calls; that attempt is excluded from these per-case means and disclosed in the amendment. The successful run had one retried request and no scored failures.

The path-only lexical controls underperformed content-and-path FTS5 at eight files. The two-search `rg` script improved fixed `rg` file recall@8 by only +0.015 [-0.061, 0.083] and lost at sixteen; its mean measured wall time was 2.67 seconds. Neither result implies a trained model or an adaptive agent would behave the same way. The flat result rules out a claim that the hierarchy itself is shown to cause the original gain over lexical search. It leaves an operational trade-off: tree routing found fewer gold files than flat Jev while using fewer calls and tokens.

4.5 Illustrative gains and failures

In a GitHub CLI issue asking to hide the current branch in `pr status --repo`, the annotated file was `pkg/cmd/pr/status/status.go`. Tree routing followed the command hierarchy and ranked it eighth; neither FTS5 nor fixed `rg` included it in the first eight. Conversely, for a VS Code issue about command links rendered from Markdown, FTS5 ranked the annotated `src/vs/base/browser/markdownRenderer.ts` fifth, while tree routing followed workbench command files and missed it. Equal-k fusion also missed the renderer in its first eight. These examples were selected after aggregate analysis to show both a lexical-vocabulary failure and a routing/fusion failure; they are not independent evidence of average performance.

5 Related work and boundaries

Hierarchy-guided retrieval is not new: PageIndex [5] applies tree reasoning to documents. Code-localization work exploits structure in other ways. LocAgent [1] uses a graph of code entities and dependencies for

multi-hop localization; repository-aware file-path retrieval [7] trains a model to predict paths from natural-language queries. These methods test different trade-offs from a folder walk. Blink [2] already navigates codebase folders with Jev, so neither hierarchical code search nor Jev use is claimed as a first here.

Lexical search is a serious baseline, not a straw man. Better Call Grep [6] studies grep-like retrieval for repository-level completion. Our narrower question is whether tree routing adds gold files to a fixed-budget lexical candidate list when issue wording and source vocabulary differ, and whether that addition survives context selection. A same-agent comparison is needed before claiming an adaptive coding workflow improves; offline list fusion alone cannot establish that. Nor do these fixed-query baselines exhaust the space of intent-aware lexical search or query reformulation.

6 Limitations and conclusion

On this cohort, name-only tree routing found annotated files omitted by both one-shot lexical rankings and improved average file and delivered-line recall over FTS5 at equal budgets. The post-study flat model path tournament did better at retrieval quality while using more calls and tokens, so the study does not establish that hierarchy causes the gain over lexical search. Tree+FTS5 improves the lexical arm but does not reliably beat tree alone; fusion can push required files out of the shortlist.

The primary study tests an issue’s first fixed query, not an adaptive agent. A multi-turn agent can reformulate searches, inspect stack traces, and follow references. Therefore a tree-only file-recall advantage over one fixed `rg` ranking would establish complementary candidate generation, not superiority over `grep` as a tool or over `Pi` as an agent. Fusion improves an agent only if the added files survive its context selection and are actually used; we measure delivered lines but not issue resolution.

The corpus policy excludes hidden files and directories commonly used for generated outputs. Five selected cases lost gold files to that rule, including real source under directories named `build`; those exclusions are reported, not silently swapped. Folder names can also be misleading, and the classifier may prune an essential branch. The pre-fix gold text alignment check found 27 audited cases with at least one mismatch; line-level results on those cases are less secure than file-level results. The 11-repository convenience cohort and one provider/model may not generalize to other repository layouts or query styles. The original larger tree run was shortened after interim outcomes were visible, although the reduced-case rule was independent of those outcomes; the resulting evidence is weaker than a fully blinded preregistered replication. The post-study flat Jev tournament used a different instruction and more calls than tree. A call- and prompt-matched comparison with repeated stochastic runs would be needed to estimate the hierarchy’s effect independently of compute allocation. A chat-model gate would test the role of the Choice endpoint. The scripted two-search `rg` control does not cover agent-led query revision; code-symbol indexes and downstream coding agents remain untested. Provider prices and latency can change, so the paper reports observed calls/tokens and setup costs alongside dollar estimates.

The practical implication is to evaluate directory routing as an additional *file-candidate source*, especially when one-shot lexical query terms scatter across a repository. Its 8.93-second mean latency and model charge make a blanket replacement for FTS5 unattractive; future systems should test when to invoke it, how to preserve complementary candidates, and whether an agent actually uses the extra evidence. This study establishes that added evidence on a fixed benchmark, not a measured improvement in software repair.

7 Reproducibility and data availability

The code and data are at github.com/manojbajaj95/fastindex, commit 50721e1. That stable commit records the case selection, source policy, parameters, budget amendments, and reproduction commands under `paper/` and `evals/`. It includes the pinned case manifest, a ledger of all 82 retained case IDs, commits, query hashes, gold ranges, and five exclusions, plus copies of the non-secret tree, FTS5, `rg`, flat Jev, path-only and feedback controls, gold-text audit, and original and post-study scored-result artifacts under `paper/data/`. Their SHA-256 checksums are in `paper/data/README.md`. The audited holdout and source snapshots can be regenerated from the pinned public commits. No API credentials or repository snapshots are part of the paper artifact. The archived rankings support exact re-scoring without model calls; raw Choice responses were not retained, so they cannot be reconstructed from the ranked paths. A fresh model run may choose different paths. The study archive therefore supports analysis reproduction, not deterministic model replay. From a fresh checkout with `uv`, `git`, and `curl`, run `sh paper/reproduce.sh`. This checks the dataset checksum, materializes pinned source snapshots, and re-scores the archived rankings without API credentials. The script contains the exact stage commands and writes local results under `evals/results/`.

AI assistance and responsibility. OpenAI Codex assisted with implementation of the evaluation code, execution and analysis of the experiments, and preparation of this manuscript. The author is responsible for the study design, artifact checks, interpretation, and final text. Codex was a research and writing tool; the routing arm evaluated TypeSafe Jev 1.13.

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